

TESLA STOCK PRICE PREDICTION

Deep Learning Based Stock Forecasting using SimpleRNN
& LSTM

Prepared By

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1. Introduction

This project focuses on predicting Tesla stock prices using Deep Learning techniques such as SimpleRNN and LSTM. Stock market prediction is a time-series forecasting problem where historical trends are analyzed to estimate future prices. The project uses Tesla historical stock data including Open, High, Low, Close, Adj Close, and Volume columns. The final system provides future stock predictions and an interactive Streamlit dashboard.

2. Problem Statement

The objective of this project is to develop a deep learning-based stock price prediction system that can forecast Tesla stock prices accurately. Traditional machine learning models often struggle with sequential data, therefore recurrent neural networks are used. The project aims to compare the performance of SimpleRNN and LSTM models for stock prediction.

3. Objectives

The primary objective is to analyze Tesla stock market trends and predict future stock prices using Deep Learning models. Additional objectives include data preprocessing, exploratory data analysis, feature engineering, model training, evaluation, and deployment using Streamlit.

4. Dataset Description

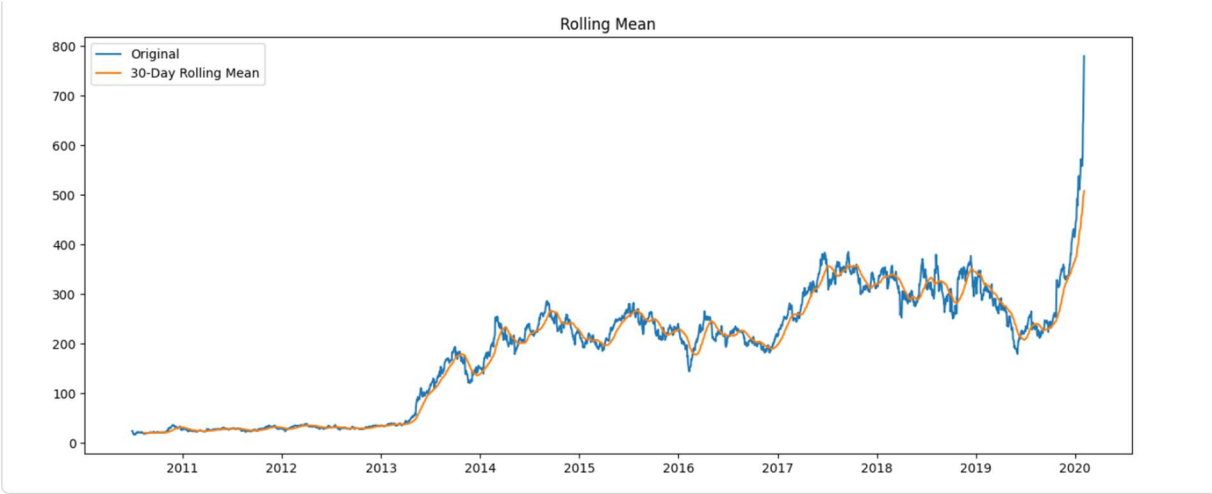
The dataset contains Tesla stock historical data with attributes such as Date, Open, High, Low, Close, Adj Close, and Volume. The Adj Close column was selected as the target variable because it reflects stock splits and dividend adjustments. The dataset was loaded using Pandas and indexed using the Date column.

5. Data Preprocessing

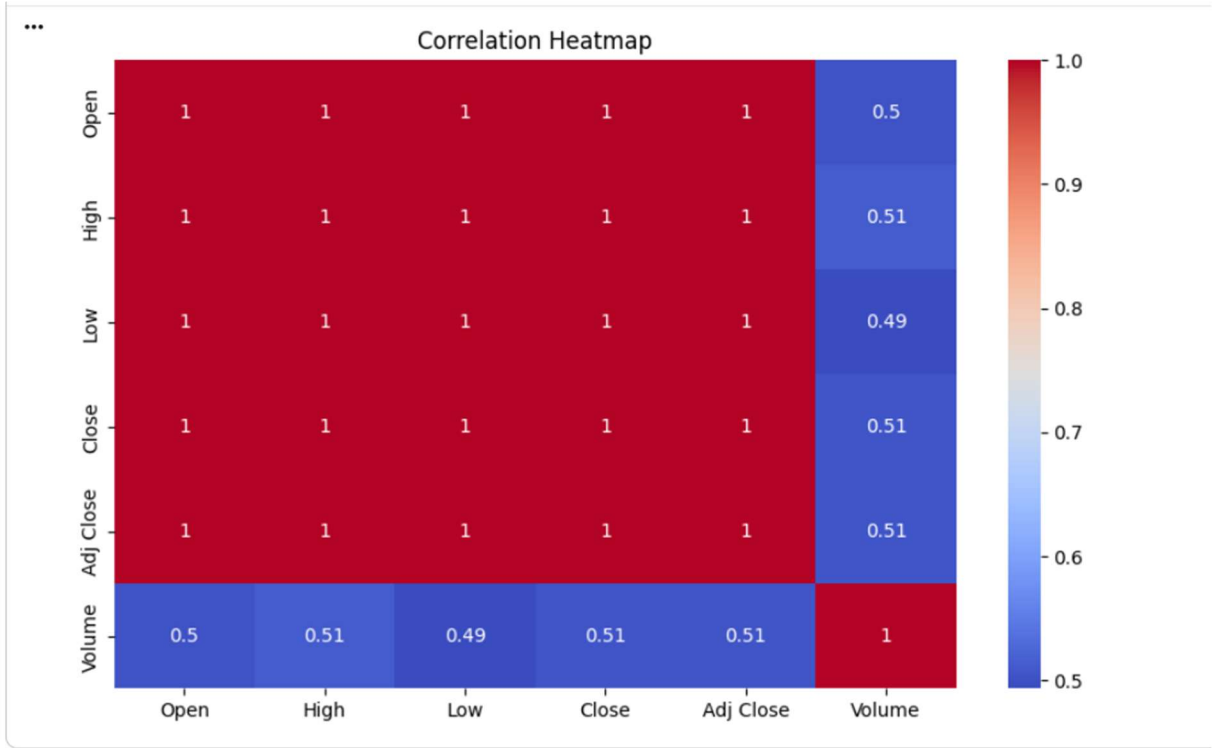
Data preprocessing included handling missing values, converting the Date column into datetime format, and scaling numerical values using MinMaxScaler. The stock price values were normalized between 0 and 1 to improve deep learning model convergence. Time-series sequences were created using a sliding window approach.

6. Exploratory Data Analysis (EDA)

EDA was performed to understand stock market trends, correlations, volatility, and moving averages. Several visualizations were created to identify trends and insights from the dataset. The charts helped understand the behavior of Tesla stock prices over time.



Rolling Mean Analysis



Correlation Heatmap

7. Feature Engineering

Feature engineering involved creating rolling averages and time-series sequences from the stock price data. The previous 60 days stock values were used to predict future stock prices. This helped the deep learning models learn sequential dependencies effectively.

8. Model Building - SimpleRNN

A SimpleRNN model was developed using TensorFlow and Keras Sequential API. The architecture included SimpleRNN layers, Dropout layers to reduce overfitting, and Dense layers for prediction output. The model was trained using Mean Squared Error loss function and Adam optimizer.

9. Model Building - LSTM

An LSTM model was implemented to capture long-term dependencies in sequential stock data. LSTM performs better than traditional RNN in handling vanishing gradient problems. The model was trained on historical Tesla stock prices and used for future forecasting.

10. Model Evaluation & Comparison

The performance of SimpleRNN and LSTM models was evaluated using MSE, RMSE, and MAE metrics. The comparison helped identify which model performed better for Tesla stock price prediction. Visualization of predicted vs actual values was also performed.

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Compare SimpleRNN vs LSTM

[]

```
comparison = pd.DataFrame({  
    'Model' : ['SimpleRNN', 'LSTM'],  
    'MSE' : [rnn_mse, lstm_mse],  
    'RMSE' : [rnn_rmse, lstm_rmse],  
    'MAE' : [rnn_mae, lstm_mae]  
})  
  
comparison
```

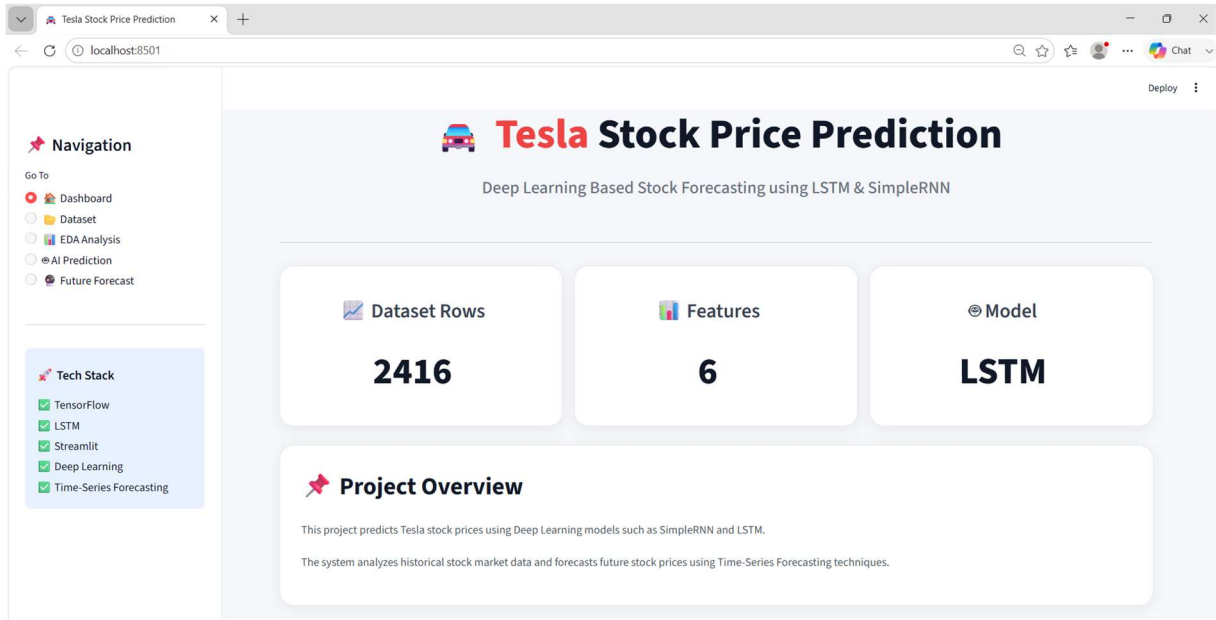
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	Model	MSE	RMSE	MAE
0	SimpleRNN	269.896850	16.428538	10.533865
1	LSTM	1099.856577	33.164086	22.896801

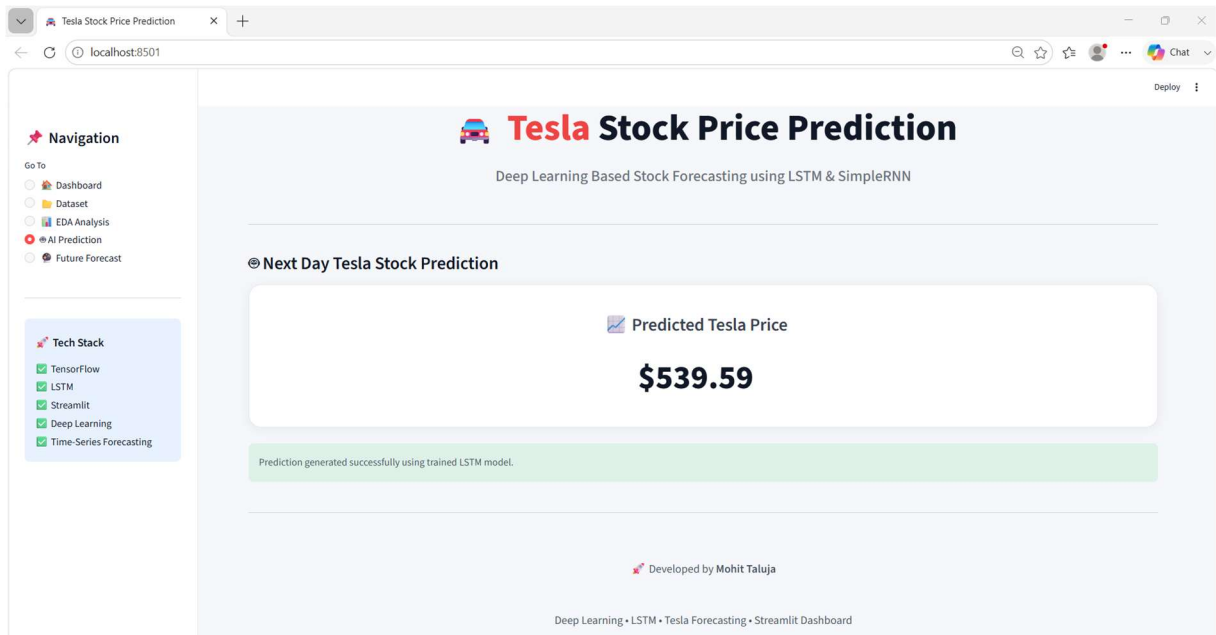
SimpleRNN vs LSTM Model Comparison

11. Streamlit Deployment

A professional Streamlit dashboard was developed for interactive stock prediction and visualization. The dashboard allows users to explore the dataset, perform EDA, and generate AI-powered stock forecasts. The UI was designed using modern layouts and professional color themes.



Professional Streamlit Dashboard



AI Prediction Interface

12. Conclusion & Future Scope

The project successfully implemented Deep Learning models for Tesla stock price prediction. The LSTM model demonstrated strong capabilities in learning sequential patterns from historical stock data. Future improvements may include integrating sentiment analysis, macroeconomic indicators, and Transformer-based models for enhanced prediction accuracy.